13.1 The Indian Cattle Leech (*Hirudinaria granulosa*)

Taxonomic Position	
Phylum	Annelida
Class	Hirudinea
Order	Gnathobdellida
Genus	*Hirudinaria*
Species	*granulosa*

13.1.1 Habit and Habitat

Hirudinaria granulosa (Indian Cattle Leech) is found in India, Bangladesh, Pakistan, Myanmar and Srilanka. It lives in freshwater ponds, lakes, swamps and slow streams. They are **ectoparasitic** and feed on the blood of fishes, frogs, cattle and human. It is **sanguivorous** (blood sucking) in nature.

13.1.2 External Morphology

Shape and Size: The body of a leech is soft, vermiform, elongated and segmented. It becomes ribbon shaped when extended and almost cylindrical when contracted. Leeches may grow to a length of 35cm.

Colouration: Dorsal surface is olive green in colour and the ventral surface is orange yellow or orange red in colour.

Segmentation: Metamerism is the segmentation of the body. The body of leech is metamerically divided into 33 **segments.** The segments are arranged one behind the other. Each segment is further superficially subdivided into *rings* or *annuli*. A temporary **clitellum** is formed on segments 9-11, which is meant to produce a **cocoon** during the breeding season.

Receptors: On the dorsal side there are five pairs of eyes on the first five segments. Each segment bears a number of sensory projections called receptors. **Annular receptors** are located in each annulus and **segmental receptors** are located on the first annulus of each segment.

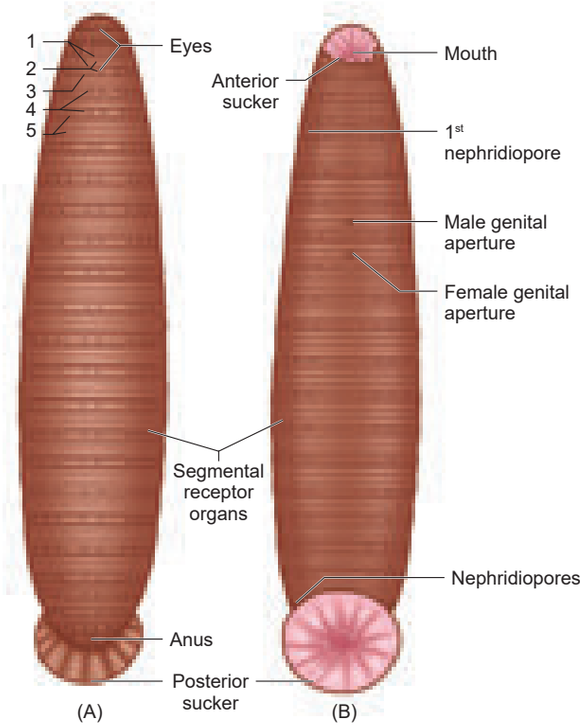


Figure 13.1 External morphology of Leech

A. Dorsal View B. Ventral view

Suckers: Leech has two suckers. The sucker located at the anterior end is called **anterior sucker** or **oral sucker** which is ventral in position occupying the first five segments. The **posterior sucker** is formed by the fusion of the last seven segments. The anterior sucker helps in feeding, while both the suckers help in attachment and locomotion.

External apertures

(i) **Mouth:** It is located in the middle of anterior sucker.

(ii) **Anus:** Anus is a small aperture that opens on the mid-dorsal side of 26th segment.

(iii) **Nephridiopores:** Nephridia open to the exterior by 17 pairs of nephridiopores. They lie ventrally on the last annulus of each segment from 6 to 22.

(iv) **Male genital pore:** It is a mid-ventral opening, situated between second and third annuli of 10th segment.

10th_Science Unit-13.indd 188 11-01-2020 15:39:01